Ex 5: Information Extraction using spaCy

Question

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Department of Artificial Intelligence and Data Science
V Semester B. Tech AIDS
AD23532 – Principles of Data Science

Academic Year: 2026-2027 **Batch: 2024-2028**

Ex5: Information Extraction using spaCy

Learning Objective:
The objective of this experiment is to enable students to understand and implement information extraction techniques on unstructured customer review text using spaCy.

Problem Statement:
An e-commerce company receives thousands of customer reviews containing valuable information about products, brands, organizations, locations, quantities, and other entities. Manually identifying and organizing this information from large volumes of reviews is time-consuming and inefficient.

As an NLP Engineer, your task is to process customer reviews using the spaCy Natural Language Processing library and develop an information extraction system that identifies and extracts meaningful linguistic information and named entities from the reviews. The extracted information should be organized into a structured format that can be further used for customer feedback analysis.

Tasks to Perform:

1. Reuse the Amazon Fine Food Reviews Dataset used in the previous experiment and load it into a Pandas DataFrame. Install and import the required libraries.

Clean the dataset

2. Select the Text column containing customer reviews, Remove missing or null values, Remove duplicate reviews (if any), Retain only the first 10,000 reviews for faster processing.

Text Preprocessing and Part-of-speech (POS) tags

3. Apply the spaCy pipeline to the review text. For any five reviews, display: Original review, Tokens, Lemmas, Part-of-speech (POS) tags.
4. Identify and display different POS categories such as: Nouns, Verbs, Adjectives, Adverbs.
5. Count and display the frequency of the major POS categories present in the reviews.

Named Entity Recognition (NER)

6. Apply Named Entity Recognition (NER) using spaCy to identify entities in the reviews. Extract and display entities such as: PERSON, ORG, DATE, MONEY, QUANTITY, PRODUCT, where recognized by the selected spaCy model. Display the extracted information in a structured DataFrame.

Information Extraction

7. Create a function that accepts a customer review as input and returns: Tokens, Lemmas, POS tags, Named entities, Entity labels.
8. Test the function using at least three new customer reviews.
9. Display the extracted information in a structured format.

Aim

To extract linguistic information (tokens, lemmas, POS tags) and named entities from customer reviews using spaCy and organise it in a structured format.

Procedure

1. Load the Amazon Fine Food Reviews dataset; install and import pandas, spaCy and collections.Counter; download en_core_web_sm.
2. Select the Text column, remove nulls and duplicates and retain the first 10,000 reviews.
3. Apply the spaCy pipeline and display the original text, tokens, lemmas and POS tags for five reviews.
4. Identify nouns, verbs, adjectives and adverbs, and count the frequency of the major POS categories.
5. Run Named Entity Recognition and tabulate entities (PERSON, ORG, DATE, MONEY, QUANTITY, PRODUCT) in a DataFrame.
6. Create an extraction function returning tokens, lemmas, POS tags, entities and labels; test it on three new reviews.

Code

```
#Mohammed Fayyad(241801165)
# !pip install spacy pandas && python -m spacy download en_core_web_sm
import pandas as pd, spacy
from collections import Counter

nlp = spacy.load("en_core_web_sm")
df = pd.read_csv("Reviews.csv")
df = df[["Text"]].dropna().drop_duplicates().head(10000).reset_index(drop=True)

# Five reviews: tokens, lemmas, POS
for text in df["Text"].head(5):
    doc = nlp(text)
    print("Original:", text)
    print("Tokens  :", [t.text for t in doc])
    print("Lemmas  :", [t.lemma_ for t in doc])
    print("POS     :", [(t.text, t.pos_) for t in doc]); print()

# POS categories and frequencies (first 1000 reviews for speed)
pos_count = Counter(); groups = {"NOUN": [], "VERB": [], "ADJ": [], "ADV": []}
for doc in nlp.pipe(df["Text"].head(1000), batch_size=50):
    for t in doc:
        pos_count[t.pos_] += 1
        if t.pos_ in groups: groups[t.pos_].append(t.text.lower())
for k, v in groups.items():
    print(k, Counter(v).most_common(10))
print(pos_count.most_common())
```

```
# Named entities
wanted = {"PERSON", "ORG", "DATE", "MONEY", "QUANTITY", "PRODUCT"}
rows = []
for i, doc in enumerate(nlp.pipe(df["Text"].head(1000), batch_size=50)):
    for ent in doc.ents:
        if ent.label_ in wanted:
            rows.append([i, ent.text, ent.label_])
ent_df = pd.DataFrame(rows, columns=["Review_ID", "Entity", "Label"])
print(ent_df.head(20)); print(ent_df["Label"].value_counts())

# Extraction function
def extract_info(review):
    doc = nlp(review)
    return {
        "Tokens": [t.text for t in doc],
        "Lemmas": [t.lemma_ for t in doc],
        "POS": [(t.text, t.pos_) for t in doc],
        "Entities": [e.text for e in doc.ents],
        "Entity_Labels": [(e.text, e.label_) for e in doc.ents],
    }

tests = [
    "I paid $12 for two pounds of Starbucks coffee in March and it was great.",
    "Nestle chocolate from Switzerland arrived on 5 May, but the box was damaged.",
    "John bought three bottles of Tabasco sauce at Walmart yesterday.",
]
for t in tests:
    print(pd.Series(extract_info(t))); print("-" * 60)
```

Result

Tokens, lemmas, POS tags and named entities were extracted from the reviews. POS frequencies and entity counts were tabulated, and the extraction function returned structured output for three new test reviews.

```
      Review_ID    Token    Lemma    POS
0             1        I        I   PRON
1             1     have     have    AUX
2             1   bought      buy   VERB
3             1  several  several    ADJ
4             1       of       of    ADP
..          ...      ...      ...    ...
280           5     this     this   PRON
281           5       is       be    AUX
282           5        a        a    DET
283           5     deal     deal   NOUN
284           5        .        .  PUNCT

[285 rows x 4 columns]
     Review_ID                   Entity         Label
0            1                 Vitality           ORG
1            1              My Labrador        PERSON
2            2             Jumbo Salted           ORG
3            2                    Jumbo  WORK_OF_ART
4            3  around a few centuries          DATE
5            3  pillowy citrus gelatin           ORG
6            3                 Filberts           GPE
7            3             C.S. Lewis'           ORG
8            3     The Lion, The Witch  WORK_OF_ART
9            3                  Edmund       PRODUCT
10           3                 Sisters        PERSON
11           3                   Witch           ORG
12           4              Robitussin           GPE
13           4  the Root Beer Extract I           LAW
14           5                Delivery        PERSON
```